

White by Construction: Two Crypto Stat-Arbitrage Screens That Pass Their Own Placebos

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Abstract

We set out to build a profitable statistical-arbitrage strategy on hourly cryptocurrency pair spreads. We report mostly what does not work, and why that is the more useful result. Two screens that appear to find strong, tradable structure are statistical artifacts. A Kalman dynamic hedge ratio appears to produce stationary, mean-reverting spreads for nearly the entire liquid universe, but it runs a stationarity test on a filter’s own innovations, which are white by construction; placebos that cannot be cointegrated pass the same screen at the same near-100% rate, while a clean test finds genuine out-of-sample cointegration in only a few percent of pairs. A rolling z -score appears to generate a large reversion edge, but it reverts mechanically even on a pure random walk. Both artifacts hold at hourly, four-hour, and daily frequencies. Once we benchmark against placebos, the structure that survives is either priced too fast to trade or too marginal to deploy: order-flow information in the Level-2 book decays within seconds, reversion speed is a selectable property of pairs (Spearman $\rho = 0.46$) but selectability is not profit, and the one market-neutral mean-reversion book that clears honest accounting posts a HAC- and venue-adjusted monthly Sharpe near 1.0, and only as a never-stop, multi-week hold. We read the result as a market-efficiency statement about crypto pair spreads, and as a cautionary standard: a cointegration or z -score screen should not be trusted until it has been shown to fail on a matched placebo.

1 Introduction

We set out to build a profitable statistical-arbitrage strategy on hourly cryptocurrency pair spreads. We did not. The more useful finding is why: the two screens that made the strategy look best are statistical artifacts, and the structure that is genuinely present is either priced within seconds or too marginal to trade after costs and selection.

The project began in the pairs-trading tradition of Gatev, Goetzmann, and Rouwenhorst [10]: find two assets whose prices hold a stable relationship, trade the spread when it deviates, and collect the reversion. Cryptocurrency is a natural place to look, with a continuously traded market and a fast-changing set of listed coins. Our proposal was non-committal about profit from the start. It framed the goal as quantifying whether stable mean-reverting relationships exist in crypto, “as a way of gaining a deeper understanding of market efficiency and information flow.” This paper answers that question, and the answer is mostly negative.

We make three contributions. First, two screens in common use manufacture the appearance of structure. A Kalman dynamic-hedge cointegration test reports out-of-sample cointegration in almost every liquid pair, but it tests a filter’s own innovations, which are near-white by construction; placebos that cannot be cointegrated pass at the same near-100% rate, at hourly, four-hour, and daily frequencies. A rolling z -score reports strong mean reversion, yet it reverts mechanically even

on a pure random walk. Each is exposed by a single matched placebo, and the placebo is the deliverable. Second, the structure that survives is not tradable: order flow in the Level-2 book carries real information with a seconds-scale half-life, and reversion speed is a selectable property of pairs that does not convert into profit. Third, one market-neutral mean-reversion strategy clears honest accounting at a monthly Sharpe near 1.0, but only without a stop and only as a patient multi-week hold, and a pre-registered held-out backtest of it inflates to a Sharpe near 5 on micro-cap moves no one could trade. Even the survivor illustrates the thesis.

The methodological lesson is narrow and reusable: a cointegration or mean-reversion screen should not be believed until it has been shown to fail on a matched placebo. The backtest-overfitting literature [2, 12] checks the strategy that gets selected; we check the screen that builds the spread.

2 Related work

Our screens come from the pairs-trading and statistical-arbitrage literature. The distance and cointegration approaches of Gatev et al. [10] and Vidyamurthy [22], the residual mean-reversion framework of Avellaneda and Lee [1], and the taxonomy in Krauss [16] (with machine-learning variants in [17]) define what we test. Our dynamic hedge is the state-space pairs model of Elliott et al. [7], built on the cointegration of Engle and Granger [8]. The contribution here is not a new screen but a placebo that two standard ones fail.

For cryptocurrency the efficiency question is unsettled. Urquhart [21] finds early Bitcoin weak-form inefficient, while adaptive-markets evidence [20, 15, 19] finds efficiency time-varying and trending up. The closest prior on crypto pairs trading, Fil and Kristoufek [9], reports strategies that underperform benchmarks and are highly cost-sensitive. Our results agree and sharpen the mechanism: the apparent edge is either a screening artifact or priced within seconds.

The microstructure tests use standard tools: Kyle’s λ [18], order-flow imbalance [4], VPIN [6], PIN [5], and information shares [13]. The placebo logic is the backtest-overfitting literature applied one level earlier. Bailey and coauthors show a handful of trials manufacture spurious in-sample performance [2, 3]; Harvey et al. [12] and Hou et al. [14] show most published factors fail multiple-testing correction; and Gelman and Loken [11] show data-contingent choices inflate error even without explicit fishing. We apply that skepticism to the screens that build the spread, not only to the strategy selected on it.

3 Data and market setting

The spread work uses hourly OHLCV for USDT spot pairs from the Binance.US API, stored locally, spanning 2021 through May 2026. The cross-exchange check pulls the same hours from Coinbase (USD-quoted) for the overlapping majors. The microstructure work uses event-level Level-2 data from Tardis: top-25 book snapshots and the trade tape for Binance spot, the 50 most liquid USDT symbols, over 2024. We keep these separate on purpose. The spread and reversion results are Binance.US and Coinbase hourly klines; only the microstructure results use the Tardis book.

The investable universe is the most liquid USDT symbols, constructed as of each analysis time. A symbol is eligible only if its history begins before the analysis window and reaches its start, a point-in-time rule added after an audit found that later listings could otherwise leak into the historical universe. The local store holds currently listed symbols, so delisted coins are mostly absent; we treat this as a survivorship limitation (Section 7), not a solved problem. Costs are

charged on every position change at 15 bps per leg (10 bps fee, 5 bps slippage), about 30 bps round-trip for the two-leg spread, and stated before any performance number.

Three rules hold throughout. Inference uses Newey–West HAC standard errors, because overlapping targets and month-long holds make naive errors far too small; in one case a spurious five-minute reversal collapsed from $t = -22.8$ to $t = -1.4$ under HAC. Evaluation is strict walk-forward, six-month train and three-month test, with hedge ratios and pair selection fit on the train window only. And every positive claim is checked against a matched placebo, a random-walk, phase-randomized, block-shuffled, or random-pair null built to reproduce the claim if it is mechanical. Selection across many configurations is corrected with a deflated Sharpe ratio [2].

For the one trading result that survives (Section 6) we pre-registered a single configuration: the parameters and the held-out evaluation window were written to a file and committed before the run, then evaluated once and reported whatever the result was. The lock and its one pre-run correction are in the repository’s commit history.

4 Two screening artifacts

Two screens drove our early results, and both are artifacts. The first, a Kalman dynamic hedge ratio, appeared to find out-of-sample cointegration in almost every liquid pair. The second, a rolling z -score, appeared to find strong mean reversion. We show that each passes a placebo it should fail, and that neither carries the information it seems to. This is the paper’s main methodological contribution. Both screens are in common use, and a careful analyst is easily fooled by either one.

4.1 The Kalman dynamic-hedge cointegration screen

A static hedge ratio fits one β for the whole sample. A Kalman dynamic hedge lets β drift as a random walk and updates it each step from the data. We fit the process and observation noise by maximum likelihood on the training window, froze them, and rolled the filter forward on the test window. The test-period innovations, the filter’s one-step prediction errors, became the spread, and we ran an augmented Dickey–Fuller (ADF) test on them. Under this screen 100% of the top-50 pairs rejected the unit root out of sample at $p < 0.05$, and 96.4% at $p < 0.001$. A static-OLS spread over the same window passed only about 3% of pairs. The contrast looked decisive.

It is decisive about the filter, not the pair. A Kalman filter with non-zero process noise adapts its hedge ratio to track whatever relationship is present. Its innovations are then driven toward whiteness, exactly white only for a correctly specified filter at steady state, and approximately so here on the frozen-parameter test window. An ADF test on a near-white series rejects the unit root almost always, whether or not the two assets are cointegrated. Testing the stationarity of a filter’s own innovations is circular. An earlier draft of this project defended the result on the grounds that the parameters were fit on training data only and never refit on test. That defense misses the mechanism. The whitening holds out of sample too, because the filter keeps adapting on the test window from its frozen dynamics.

A placebo settles it. We ran the identical screen on series that cannot be cointegrated: (a) independent simulated random walks, (b) each coin paired with a phase-randomized surrogate of another coin, and (c) real pairs with one leg block-shuffled. All three pass at the same near-100% rate as the real pairs (Table 1). A clean, non-circular test tells the opposite story. Engle–Granger [8] on the test window, and a static-OLS ADF, put genuine out-of-sample cointegration at 2–3% of pairs on the matched window, at or below the 5% null floor. The screen carries no cointegration information. That is not a claim that no pair ever co-moves; it is that the screen cannot tell. Its 100% is noise dressed as signal.

Table 1: The Kalman screen passes placebos that cannot be cointegrated at the same rate as real pairs. Out-of-sample ADF on the Kalman innovations, top-50 USDT universe (250-pair subsample), $t_0 = 2024-01-01$, 90-day train / 30-day test.

Series tested	ADF $p < 0.05$	ADF $p < 0.001$
Real top-50 pairs	100.0%	96.4%
(a) Independent random walks	100.0%	98.3%
(b) Coin vs. phase-randomized surrogate	100.0%	95.0%
(c) Real pair, one leg block-shuffled	100.0%	95.0%
Clean tests (EG and static OLS, test window)	2–3%	—

The artifact does not depend on the bar. We reran the screen at hourly, four-hour, and daily frequencies on a recent multi-year window (Table 2). Real pairs and random-walk placebos pass together at every cadence (Figure 1): 100% versus 100% at hourly and four-hour, and 70% versus 68% at daily ($p < 0.001$). The clean tests behave differently. Engle–Granger runs at 25% on this long hourly window and falls to the 5% null floor at daily; a static-OLS ADF runs higher still, from 42% at hourly to 8% at daily. Both are window-dependent. Over long hourly spans the crypto majors share a common trend, which a clean test reads as broad co-movement, and which the matched short window of Table 1 puts back at the 2–3% floor. Neither clean test approaches the screen’s 100%, and the small difference between the two tables’ real $p < 0.001$ rates (96% versus 100%) is window and sample size, not a contradiction. The Kalman screen carries no cointegration information at any frequency we tested.

Table 2: Frequency robustness of the Kalman artifact. Real pairs and random-walk placebos pass the screen together at every cadence; the clean Engle–Granger and static-OLS benchmarks do not. Top-50 universe, 60 pairs per cell, recent multi-year window.

Frequency	Real pairs		RW placebo		Clean tests $p < 0.05$	
	$p < 0.05$	$p < 0.001$	$p < 0.05$	$p < 0.001$	Engle–Granger	static OLS
Hourly	100.0%	100.0%	100.0%	100.0%	25.0%	41.7%
4-hour	100.0%	100.0%	100.0%	100.0%	11.7%	20.0%
Daily	100.0%	70.0%	100.0%	68.3%	5.0%	8.3%

4.2 The rolling z -score

The second screen is a rolling z -score, the standard entry signal for pair trades. For a spread s_t it subtracts a trailing mean and divides by a trailing standard deviation,

$$z_t = \frac{s_t - \bar{s}_{t-1}}{\sigma_{t-1}},$$

and it trades when $|z_t|$ is large, betting the spread returns to its mean. On our spreads it produced a large reversion edge.

The edge is mechanical. Subtracting a trailing mean from any series, including a pure random walk, manufactures mean reversion, because the trailing mean chases the level and the deviation from it cannot wander as freely as the raw series. A random walk has no mean to revert to, yet a rolling z -score on one still reverts.

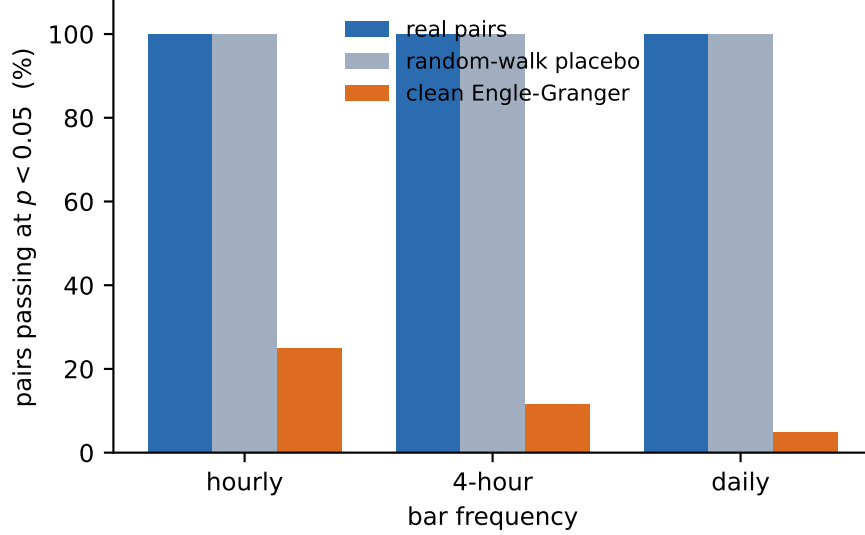


Figure 1: The Kalman screen carries no cointegration information at any frequency. Real pairs and random-walk placebos pass at the same near-100% rate at hourly, four-hour, and daily bars, while a clean Engle–Granger test stays far lower. Exact values in Tables 1 and 2.

We measured the mechanical floor two ways, and both are large. Per $|z| > 2$ event, the signed reversion on pure random walks is positive at every frequency we tested: $+0.98z$ at hourly, $+1.61z$ at four-hour, and $+1.51z$ at daily, each many standard deviations from zero (Table 3; Figure 2 shows the mechanism on one path). As a trading rule at hourly frequency, a rolling- z strategy that posts a Sharpe near 4.75 on our data still posts about 2.2 on random pairs and 2.4 on phase-randomized noise. Most of the apparent edge is the floor. For this reason every per-event reversion magnitude elsewhere in this paper is reported net of a per-pair, per-window random-walk floor.

Table 3: The rolling z -score manufactures mean reversion on pure random walks, at every frequency. Mean signed reversion per $|z| > 2$ event, 200 random-walk surrogates per cell.

Frequency	Mean event reversion (z)	s.d.
Hourly	+0.98	0.26
4-hour	+1.61	0.39
Daily	+1.51	0.45

Both screens share a failure mode. They manufacture the appearance of structure that a matched placebo reproduces. A cointegration screen built on a filter’s own innovations must be checked against a white-noise placebo before it is believed, and a rolling- z reversion claim must be checked against a random-walk null. Neither check is standard practice, and without them both screens report a signal that is not there. This is the lesson the backtest-overfitting literature draws for strategy selection [2, 12], applied one level earlier, to the screens that build the spread.

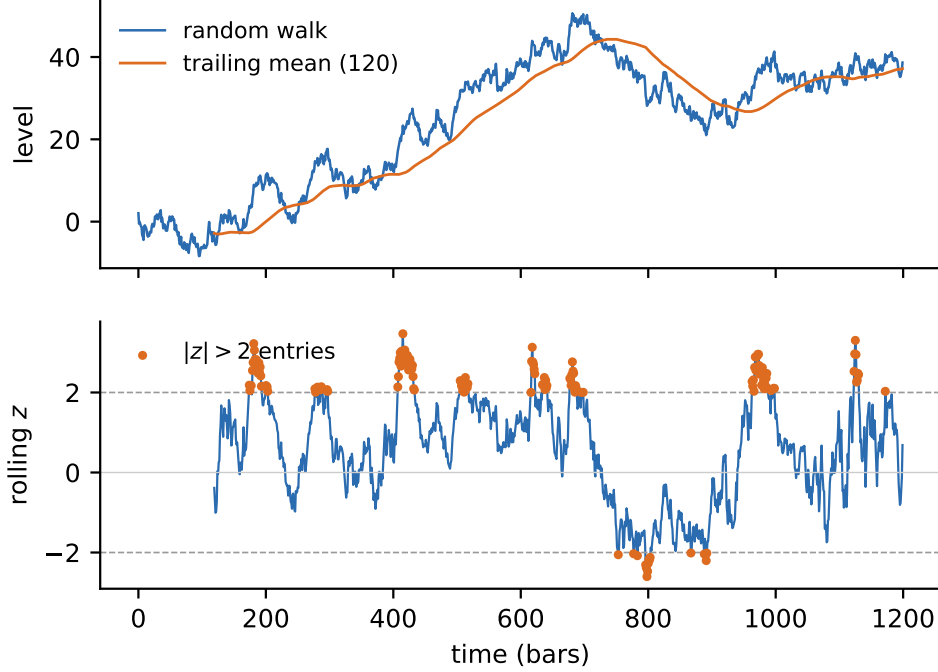


Figure 2: A rolling z -score manufactures mean reversion on a pure random walk. Top: one simulated random walk and its trailing mean. Bottom: the resulting rolling z , which crosses ± 2 and reverts toward zero, although the series has no mean to revert to.

5 What is real but not tradable

Strip the two artifacts and some genuine structure remains. None of it is tradable, for one of two reasons: it is priced faster than anyone could act, or it is a statistical property that does not convert into profit. We take the two clearest cases.

5.1 Reversion speed ranks pairs but does not pay

Mean reversion persists out of sample even though cointegration does not. Across nine disjoint walk-forward splits (six-month train, three-month test), a pair’s in-sample OU reversion speed predicts its out-of-sample *excess* reversion, net of the rolling- z floor of Section 4.2: Spearman $\rho = 0.46$ (95% CI $[0.37, 0.54]$, $p < 0.001$), positive in every one of the nine splits. The top quintile shows about $0.69z$ more excess reversion than the bottom out of sample $([0.58, 0.81], p < 0.001)$. The placebos are clean: a random quintile split gives $+0.0001z$, and shuffling the train-to-test link collapses ρ to -0.002 . Reversion speed is a real, graded, persistent property that ranks pairs.

One caveat belongs on that coefficient. The $\rho = 0.46$ is the strongest of four in-sample reversion metrics we tried, chosen after the fact, and it is not corrected for that choice. The direction holds across all nine splits; the exact coefficient is a best case.

Selectability is not profit. Ranking pairs by reversion speed tells you where reversion lives. Whether it can be extracted after costs is a separate question, and the strategy built on the property (Section 6) is marginal at best.

5.2 Order flow is informative, but priced within seconds

We tested the order-flow ideas on a 2024 sample of Binance Level-2 book and trade data: aggregate book metrics on the top-50 USDT majors, and event-level tests on four majors (BTC, ETH, SOL, AVAX) across days drawn from all twelve months, from Tardis. The book carries real, contemporaneous information about price formation. It carries none at any horizon you could trade against, an efficiency null for both forecasting and execution.

Trade size does proxy information. Large trades carry about twice the one-second price impact of small ones, and the ranking replicates across all of 2024 on four coins (one-second impact for BTC of 0.33/0.23/0.13 bps for institutional, mid, and retail, with institutional 2.0 to 2.6 $\times$ retail; the longer-horizon permanent estimate is noisier). Book-side order-flow imbalance explains within-second mid-price moves better than trade flow does (incremental R^2 of +0.12 to +0.16 for BTC and ETH), and 81 to 85% of the liquidity that leaves the best quote is cancellations, invisible to trade-only analysis.

All of it decays within seconds. The predictive content of order flow falls to near zero by 30 seconds, and at a 15-minute to one-hour horizon institutional flow predicts a mild reversal, not continuation, with an incremental R^2 near 0.03%. Hidden and iceberg liquidity is measurable (about 1 to 4% of at-best volume, with iceberg-episode rates higher) but carries no edge.

We then measured whether the signal helps execution, the usual fallback when a signal does not forecast returns. It does not (Table 4). Across 18,432 simulated parent orders over event-level book and tape, aggressive crossing is the cheapest method for these majors, a signal-timed passive rule does worse than a random one at the same posting rate, and the signal’s correlation with the per-order advantage of posting over crossing is +0.06, noise. A perfect-foresight oracle would save 1.6 bps. The opportunity is real. No contemporaneous book feature forecasts it. The null holds across all twelve months of 2024, including the August 5 crash.

Table 4: Order-flow signals do not improve execution. Implementation shortfall versus arrival mid, pooled, 30-second horizon; lower is cheaper. 18,432 orders, BTC/ETH/SOL/AVAX.

Execution rule	Cost vs. arrival mid (bps)
Aggressive (always cross)	+1.34
Naive passive (always post)	+2.79
Signal-timed (post/cross on the L2 signal)	+1.98
Random at the same posting rate	+1.88
Oracle (perfect foresight)	−0.23

The one useful byproduct is a cost number. Walking the real book puts execution costs about 17 to 23% below the flat 5 bps slippage component the earlier backtests assumed. Cheaper costs do not rescue the strategy.

6 The marginal survivor

One strategy survives, and its honest account is its bounds. There is a real, market-neutral, diversified mean-reversion effect in crypto pair spreads. Its tradable magnitude is uncertain, it exists only without a stop, and a literal backtest of it inflates spectacularly. We lead with the bounds because they are the result.

6.1 The effect is real and market-neutral

Selecting pairs by reversion speed and holding the spread to convergence, with a static train-window hedge and z -score (not the rolling z of Section 4.2), produces a positive, market-neutral return. It is diversified: beta to an equal-weight crypto index is about -0.06 , and the top ten principal-component portfolios explain only 1.7% of its return, so it is not a hidden factor bet. It replicates on an independent exchange (Coinbase, USD-quoted) and under four from-scratch reimplementations that share none of our pipeline, none of which found look-ahead, so the effect is not a coding artifact.

6.2 It exists only without a stop

The exit rule decides the sign of the P&L. With a conventional $|z| = 4$ stop the strategy loses (Sharpe -2.25), and the stop is the cause: it realizes losses on spreads that then revert. Remove the stop and hold to convergence and the strategy turns positive. The project set out to build a tightly risk-managed hourly strategy; this is the opposite, a patient never-stop book with a median hold near 35 days, in which 78% of positions do not converge within a quarter and about a third of the return is a generic survivor co-movement floor.

6.3 The magnitude is uncertain, and a literal backtest inflates it

The naive Binance Sharpe was first reported near 2.5, and four independent reimplementations put it at 2.0 to 2.2. Neither is the honest figure. Month-long holds make hourly returns heavily autocorrelated, so a Newey–West correction drops the monthly Binance number to about 1.4, and on the independent Coinbase venue it is about 0.85 to 0.90. The HAC- and venue-honest figure is near 1.0, and the edge concentrates in recent, high-volatility periods.

A pre-registered, held-out test makes the fragility concrete. We locked the configuration before running (Section 3 records the lock) and evaluated it once on held-out data, 2025-06 to 2026-05. The literal result was a monthly Sharpe near 5, *higher* than the in-sample figure, and it is not a tradable edge. It is broad-based inflation across high-volatility micro-cap alts: individual pair-windows post $+300$ to $+520\%$ in three months under a flat 15 bps cost that badly understates the true cost of trading those names, with unit-notional accounting that ignores capacity. A $+517\%$ three-month leg implies roughly a $150\times$ gross return, not realizable at any size. The same machinery on a venue without those micro-caps returns the ~ 1.0 figure. So the held-out test does not confirm an alpha; it shows the raw backtest is untrustworthy, the paper’s thesis turned on our own surviving result.

6.4 What would have to be true

The effect is also selection-sensitive and survivorship-fragile. A deflated-Sharpe correction (Section 3) survives the no-stop family of configurations but fails the whole stop-versus-no-stop search, and the supports we might cite for it (the t -statistic, the $\rho = 0.46$ of Section 5, the market neutrality) share the same universe, selection, and windows, so they are correlated rather than independent confirmations. Its survivorship robustness rests on the selector avoiding coins that later delist, not on surviving them: a held position in a coin that collapses is a -100% leg. A structural-break circuit breaker that halts a pair for the rest of the window after a single leg moves more than 50% against the position caps that tail at little cost to the Sharpe; the genuinely point-in-time test, on a universe that forces a held collapse, remains open.

The defensible claim is therefore the sensitivity itself. There is a real, market-neutral reversion signal in crypto pair spreads that can be selected by reversion speed. Whether it is worth anything

after realistic costs, capacity, and selection is unresolved, and the most honest single number, a monthly Sharpe near 1.0, is an estimate that a literal backtest does not cap from above.

7 Limitations

Five limitations bound these results, and the survivorship gap is the one most likely to change the conclusion.

- **Survivorship.** The local store holds currently listed symbols, so most delisted coins are absent. The surviving strategy’s robustness rests on the selector avoiding coins that later delist, not on surviving a held collapse; a truly point-in-time universe that forces a held delisting is the one test we have not run.
- **Selection.** No single configuration was fixed before the search and evaluated once across the whole project. The pre-registered test of Section 6 covers one configuration but cannot undo the selection that produced it, so the deflated-Sharpe caution stands.
- **Microstructure coverage.** The event-level Level-2 tests use four symbols (BTC, ETH, SOL, AVAX) over days sampled across 2024, not a continuous year on the full universe. The aggregate book metrics use the top-50; the execution and impact results are four-symbol.
- **Single venue and quote.** The spread results are Binance.US, USDT-quoted, with a Coinbase cross-check; the microstructure is Binance spot. We have not tested other quote currencies or a broader set of venues.
- **Costs and capacity.** Costs are a flat per-leg charge. We do not model market impact or capacity, which is exactly what inflates the held-out micro-cap backtest of Section 6.

8 Conclusion

The screens were the problem, not the market. The Kalman cointegration test and the rolling z -score both pass placebos they should fail, at every frequency we tested. Order flow forecasts only the next few seconds, a horizon no one can trade. The one market-neutral reversion effect that survives is real but marginal, stop-sensitive, and selection-sensitive, and a literal backtest of it inflates on costs it does not model.

The practical lesson is a standard, not a strategy. A cointegration or mean-reversion screen should be trusted only after it has been shown to fail on a matched placebo built to reproduce the claim if the claim is mechanical. For crypto the evidence points to microstructure efficiency at tradeable horizons: the information is real, and the market has already priced it.

What would move the result is a genuinely point-in-time universe with delisted coins, a capacity- and impact-aware cost model, and a single pre-registered configuration evaluated once on data the search never saw. Until then, the honest reading of crypto pair trading is the efficiency reading.

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